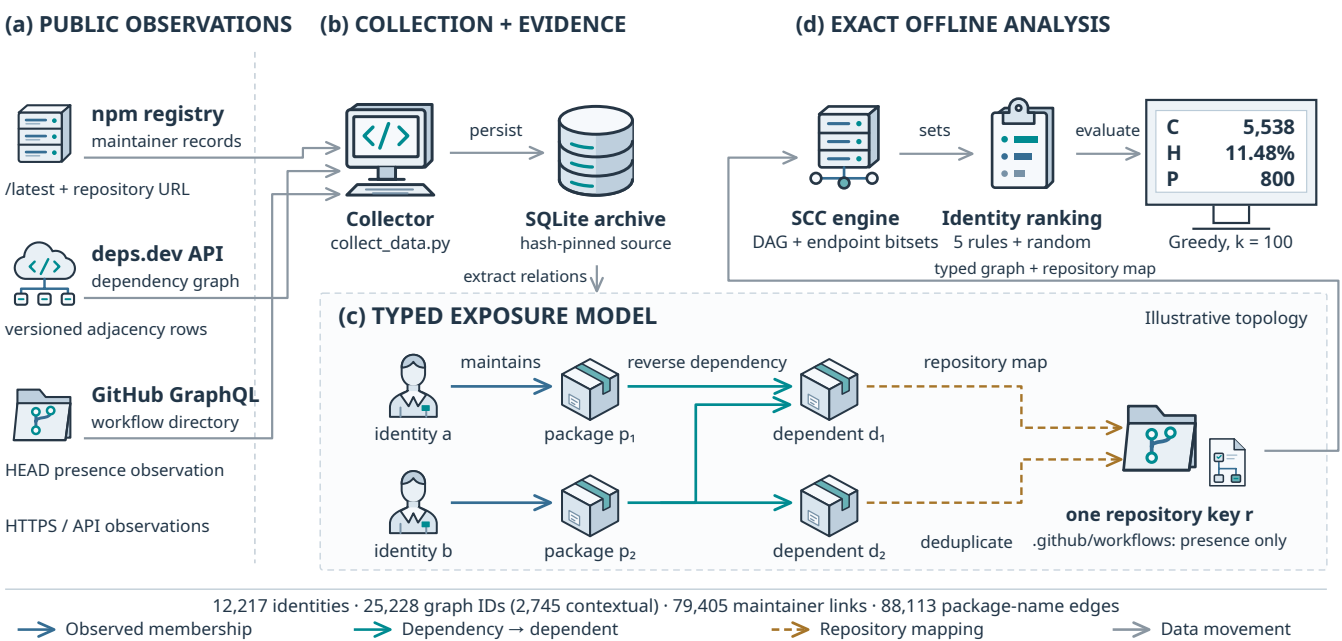
(a) PUBLIC OBSERVATIONS
(b) COLLECTION + EVIDENCE
(d) EXACT OFFLINE ANALYSIS
npm registry
maintainer records
/latest + repository URL
deps.dev API
dependency graph
versioned adjacency rows
GitHub GraphQL
workflow directory
HEAD presence observation
HTTPS / API observations
Collector
collect_data.py
persist
SQLite archive
hash-pinned source
extract relations
SCC engine
DAG + endpoint bitsets
sets
Identity ranking
5 rules + random
evaluate
C 5,538
H 11.48%
P 800
Greedy, k = 100
typed graph + repository map
(c) TYPED EXPOSURE MODEL
Illustrative topology
maintains
reverse dependency
repository map
identity a
package p1
dependent d1
identity b
package p2
dependent d2
deduplicate
one repository key r
.github/workflows: presence only
12,217 identities · 25,228 graph IDs (2,745 contextual) · 79,405 maintainer links · 88,113 package-name edges
Observed membership
Dependency → dependent
Repository mapping
Data movement